

Chiyoung Kim

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EDUCATION

Carnegie Mellon University

Visiting Scholar, Software and Societal Systems Department (S3D)

Pittsburgh, PA

Aug 2026 ~ Present

Chung-Ang University

M.S. in Intelligent Semiconductor Engineering (Expected Feb 2028)

GPA: 4.33/4.5

Selected Coursework: Intelligent Video Processing, Advanced Natural Language Processing

Seoul, South Korea

Mar 2026 ~ Present

Tecnologico de Monterrey

Exchange Student

Monterrey, Mexico

Aug 2023 ~ Jun 2024

Kookmin University

B.S. in Electrical Engineering

GPA: 3.85/4.5 (Dean's List for the final semester)

Selected Coursework: Python Programming, Computer Vision, Probability and Statistics, Linear Algebra

Mandatory military service, Republic of Korea

Seoul, South Korea

Mar 2019 ~ Feb 2026

Apr 2021 ~ Jan 2023

SKILLS

Programming & Tools: Python, PyTorch, Linux

Vision-Language-Action Models: OpenVLA, OpenVLA-OFT, CogACT, imitation learning, LoRA fine-tuning, structured pruning, hidden-state distillation

Robot Systems & Deployment: AgileX Piper manipulator, teleoperation, demonstration data collection, NVIDIA Jetson AGX Thor, on-device inference

Simulation & Benchmarking: LIBERO, LIBERO-Plus, SimplerEnv, MuJoCo

Languages: Korean (Native Speaker), English (Fluent), Spanish (Intermediate)

RESEARCH

Chung-Ang University

Graduate Research Assistant, Advisor: Prof. Minhyeok Lee

Seoul, South Korea

Mar 2026 ~ Present

- Developed an offline recovery method for structurally pruned VLA models that distills teacher hidden states without a projector, restoring a 63%-pruned OpenVLA-OFT from 0.8% to 89.7% on LIBERO-Long in about 8 GPU-hours.
- Ran a nine-ratio compression sweep on two VLA backbones and a matched width-vs-depth pruning comparison, identifying 72% reduction as the deployment operating point.
- Collected 450 teleoperated demonstrations on an AgileX PiPER arm and deployed the compressed policy on its on-board Jetson Thor, reaching 77.5% success at 2.23× lower latency and 62% less memory than the dense model.
- Co-led ReShoot, a video-diffusion pipeline that re-renders recorded demonstrations under new appearances without relabeling actions, raising recolored-object success on two real robots from 0% to 42.9% and 47.5%.
- Contributed to CoTinyVLA, a 0.9B-parameter VLA trained with chain-of-thought distillation that outperforms 7B baselines on all four LIBERO-Plus suites.

ACADEMIC PROJECTS

Carnegie Mellon University

SteelHacks Hackathon, University of Pittsburgh

- [Project description to be added after the event]

Pittsburgh, PA

Fall 2026 (Upcoming)

PUBLICATIONS

Chiyoung Kim, Sanghyuk Roy Choi, and Minhyeok Lee. "Recovering Aggressively Pruned Vision-Language-Action Models with Offline Hidden-State Distillation." *arXiv preprint arXiv:2609.19579*, 2026. (Submitted to IEEE International Conference on Robotics and Automation, ICRA 2027)

Chiyoung Kim*, Min Sung Choi*, Jinho Ju, Chanhoe Gu, Donghwan Hwang, Wonseok Choi, Woongsun Jeon, and Minhyeok Lee. "ReShoot: Generative Visual Domain Randomization of Recorded Robot Demonstrations for Visuomotor Policy Learning." *arXiv preprint arXiv:2609.19661*, 2026. (Submitted to IEEE International Conference on Robotics and Automation, ICRA 2027) *Equal contribution.

Minhyeok Lee, **Chiyoung Kim**, Chanhoe Gu, Seongrok Kim, Sanghyuk Roy Choi, Donghwan Hwang, Donghun Ryu, and Seokhyun Kim. "CoTinyVLA: Chain-of-Thought Distillation for a Sub-Billion-Parameter Vision-Language-Action Model." *arXiv preprint arXiv:2607.25487*, 2026.

TEACHING ASSISTANT EXPERIENCE

Chung-Ang University

Seoul, South Korea

Teaching Assistant – Embedded Systems Design Lab (Raspberry Pi 5)

Mar 2026 ~ Jun 2026

- Led weekly hands-on Raspberry Pi 5 lab sessions for 23 undergraduate students over a full semester, guiding hardware setup, embedded programming, and debugging.
- Revised and maintained the course lab manual for the Raspberry Pi 5 platform, distributing updated materials to enrolled students each week.
- Authored the final examination and the term-project evaluation rubric, and graded both for all 23 students.

LEADERSHIP

President, Table Tennis Club (Student Club) – Kookmin University

Sep 2019 ~ Dec 2019

- Led the university's table tennis club, organizing twice-weekly training sessions and team gatherings for members throughout the semester.
- Represented the club at an eight-university intercollegiate tournament, winning first place in lower-division doubles.

AWARDS AND HONORS

Dean's List, Kookmin University

Fall 2025

Runner-up, Lower-Div. Doubles, National Collegiate Table Tennis Tournament

2025

English Proficiency Scholarship (4 semesters), Kookmin University

2020 – 2025

Faculty Assistantship Scholarship, Kookmin University

Fall 2024

Runner-up, Lower-Div. Doubles, 8-University Table Tennis Tournament

2020

1st Place, Lower-Div. Doubles, 8-University Table Tennis Tournament

Fall 2019